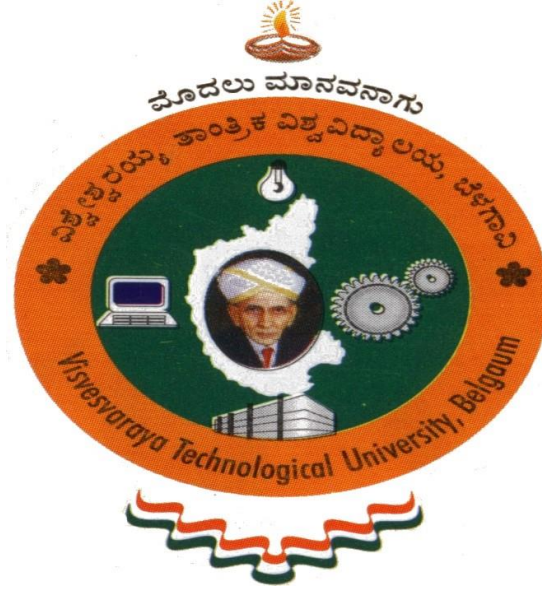


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Mini Project Report on

WIFI controlled plant disease Robot

Submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the Innovation & Design Thinking Laboratory, Department of Basic Sciences.

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CERTIFICATE

This is to certify that the mini project entitled “Automated Plant Disease Detection System” has been successfully carried out by Vikas P B , Vinayak, Shreya S, Simra and team, bonafide students of CMR Institute of Technology. The project is submitted in partial fulfillment of the requirements for the First Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the Innovation & Design patent Laboratory, Department of Basic Sciences.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library.

This mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

Signature of Guide
HOD

Raveesha KH

Signature of

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ACKNOWLEDGEMENT

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ABSTRACT

The Automated Plant Disease Detection System is designed to identify diseases in plant leaves using image processing and machine learning techniques. The main objective of this project is to help farmers and agricultural workers detect plant diseases at an early stage and reduce crop loss.

The system uses a camera module to capture images of plant leaves. These images are processed using machine learning algorithms to identify disease symptoms such as spots, discoloration, and fungal infections. The processed data is analyzed, and the detected disease is displayed to the user along with preventive suggestions.

The proposed system provides a fast, accurate, and low-cost solution for disease monitoring in agriculture. Experimental results demonstrate that the system can detect common plant diseases with good accuracy and reliability. This project serves as a practical implementation of artificial intelligence and image processing in smart agriculture.

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Chapter 1 INTRODUCTION

Agriculture plays a vital role in the economy and food production of many countries. Plant diseases are one of the major causes of crop damage and reduced agricultural productivity. Traditional disease identification methods mainly depend on manual observation by farmers or agricultural experts, which can be time-consuming and less accurate. The Automated Plant Disease Detection System is designed to provide a smart and efficient solution for identifying plant diseases using image processing and machine learning techniques. The system captures leaf images and analyzes them to detect diseases automatically.

This project aims to improve crop health monitoring, reduce manual effort, and help farmers take preventive action at an early stage.

1.2 History of Plant Disease Detection

Earlier methods of disease detection relied completely on visual inspection by experienced farmers and agricultural experts. These methods were less effective for detecting diseases at an early stage. With advancements in computer vision and artificial intelligence, automated systems have been developed for accurate and rapid disease identification.

1.3 Modern Smart Agriculture Systems

Modern agriculture systems integrate sensors, cameras, IoT devices, and machine learning algorithms for monitoring crop health. Smart farming technologies improve productivity, reduce losses, and support sustainable agriculture practices.

Chapter 2 PROBLEM STATEMENT

2.1 Description

Farmers often face difficulties in identifying plant diseases at the correct time. Delayed disease detection may lead to crop damage, reduced productivity, and financial loss. Manual

disease identification requires expert knowledge and may not always provide accurate results.

There is a need for a low-cost, reliable, and automated system that can quickly detect plant diseases and assist farmers in taking preventive measures.

2.2 Challenge Statement

How might we design a simple and cost-effective automated system that accurately detects plant diseases using image processing and machine learning techniques?

Chapter 3 DESIGN AND METHODOLOGY

3.1 Design Thinking Process

Empathize

Interaction with farmers revealed concerns about crop diseases, delayed detection, and lack of expert guidance.

Define

The main requirements identified were accurate disease detection, fast processing, and user-friendly operation.

Ideate

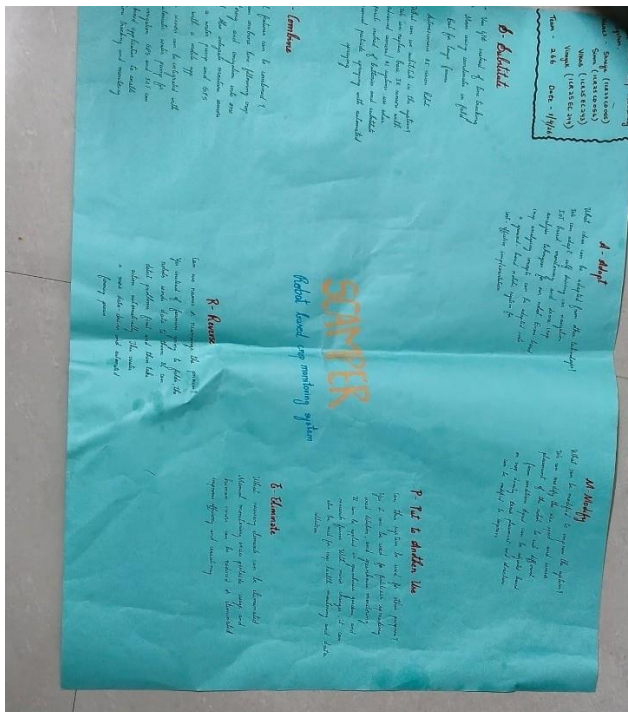
Different approaches such as manual inspection, sensor-based monitoring, and image-based disease detection were explored. Machine learning-based image processing was selected for higher accuracy.

Prototype

A prototype using a camera module, Raspberry Pi/ESP32-CAM, and machine learning model was developed.

Test

The system was tested using different leaf images, and disease detection accuracy was verified through repeated trials.



3.2 Methodology

1.The camera captures images of plant leaves.

- 2.The image is processed using image processing techniques.
- 3.The machine learning model analyzes the leaf patterns.
- 4.The disease type is identified based on trained data.
- 5.The detected disease and suggestions are displayed to the user.

3.¹ Prototype Description 3.3.1

Materials Used

- **ESP32-CAM / Raspberry Pi Acts as the main processing unit. ² Camera Module**
Captures images of plant leaves. ²

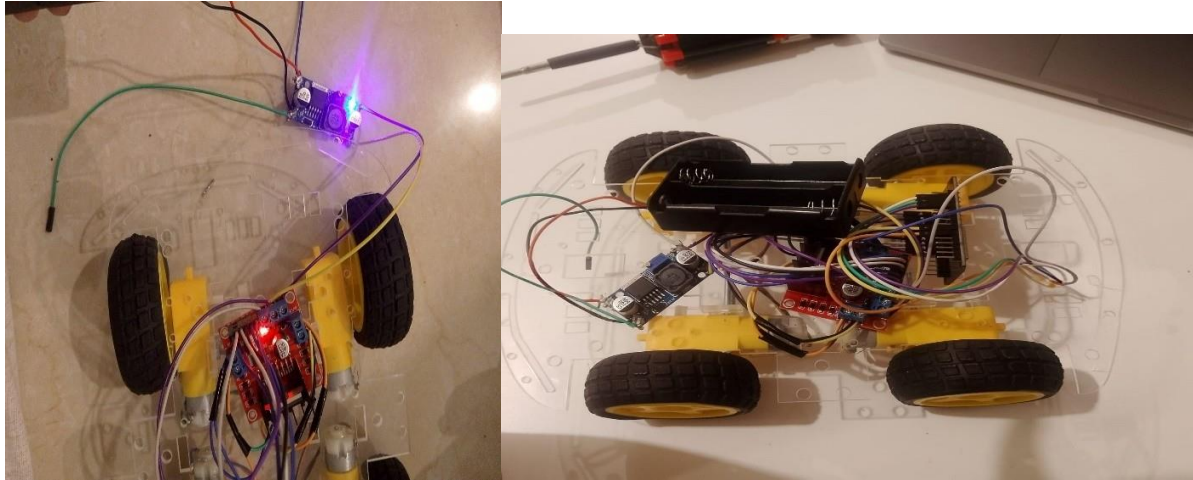
LCD Display

Displays disease information and results.

- **Power Supply**
Provides required power for the system.
- **Jumper Wires and Breadboard Used for electrical connections.**

¹ .3.2 System Architecture

The system consists of a camera connected to the processing unit. The captured images are processed using machine learning algorithms. The output is displayed on the screen with disease details and recommendations.

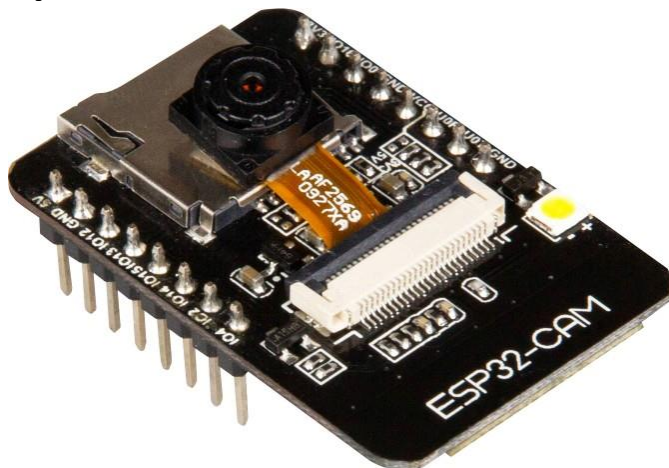


Chapter 4

IMPLEMENTATION

4.1 Hardware Implementation

The hardware setup includes ESP32-CAM/Raspberry Pi, camera module, LCD display, and power supply connections. The components were assembled and tested for stable operation.



4.2 Software Implementation

The software implementation involves programming using Python and machine learning libraries such as TensorFlow,

OpenCV, and NumPy. The model was trained using plant disease datasets.

4.3 Machine Learning Model

The machine learning model was trained using images of healthy and diseased plant leaves. Different disease classes were included to improve accuracy and performance.

Chapter 5 RESULTS AND ANALYSIS

5.1 Testing and Observation

The system was tested using multiple leaf samples under different conditions.



Quantitative Results

- **Accurate disease detection with high precision.**
- **Fast image processing and prediction time.**
- **Reliable performance for common plant diseases.**

Qualitative Observations

- **Smooth image capture and processing.**
- **Stable output results.**
- **Easy-to-use interface for farmers.**

The project results confirmed that the system successfully achieved its objectives.

Chapter 6 CONCLUSION AND FUTURE WORK

6.1 Conclusion

The Automated Plant Disease Detection System successfully demonstrates an efficient and accurate solution for identifying plant diseases using machine learning and image processing. The system helps in early disease detection, reduces crop loss, and improves agricultural productivity.

6.2 Future Work

- **Integration with IoT platforms for remote monitoring.**
 - **Mobile application support for farmers.**
 - **Real-time disease alerts and recommendations.**
 - **Cloud database integration for disease records.**
 - **Support for multiple crop types and diseases.**
-

REFERENCES

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2.TensorFlow Documentation

3.Machine Learning for Agriculture Research Papers